UNIVERSITY OF LONDON

BSc EXAMINATION 2023

For Internal Students of
Royal Holloway

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IY3660: Applications of Cryptography

Time Allowed: **TWO hours**

Answer THREE questions
No credit will be given for attempting any further questions

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Question 1 - Symmetric ciphers

(a) Give two examples of *lessons learnt* in cryptography from the cryptanalysis of historical ciphers. [4 marks]

(b) Formally define the *one-time pad* (providing message, ciphertext and key space, and the defining algorithms), and state what security it achieves.

Give two reasons why it is considered an impractical cipher. [6 marks]

(c) State and briefly describe the two main approaches to building a *block cipher*, and provide an example of block cipher for each approach.
[6 marks]

(d) *Modes of operation* are typically used to provide block ciphers with a way to encrypt flexible amounts of data.

Let E be a block cipher operating on 4-bit blocks. Let K be a key, such that encryption with E under K is given by the table below.

$0000 \mapsto 1100$	$0100 \mapsto 1001$	$1000 \mapsto 0011$	$1100 \mapsto 0100$
$0001 \mapsto 0101$	$0101 \mapsto 0000$	$1001 \mapsto 1110$	$1101 \mapsto 0111$
$0010 \mapsto 0110$	$0110 \mapsto 1010$	$1010 \mapsto 1111$	$1110 \mapsto 0001$
$0011 \mapsto 1011$	$0111 \mapsto 1101$	$1011 \mapsto 1000$	$1111 \mapsto 0010$

For example, $E_K(0000) = 1100$ and $E_K(0100) = 1001$. Likewise, the decryption of 0010 is given by $D_K(0010) = E_K^{-1}(0010) = 1111$.

Let M = $0011\ 0011$ be an 8-bit message which we would like to encrypt using the block cipher E under key K.

i. Encrypt the first block of message M with E_K using the ECB mode. [1 marks]

ii. What does the second block of message M encrypt to using the same key and the same mode of operation? [1 marks]

iii. How would you encrypt the 10-bit message M' = $0011\ 0011\ 11$ using E_K in ECB mode? [2 marks]

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iv. Encrypt the message M with E_K using the CTR mode. As counter, you should use the convention ctr_0 = 0000, ctr_1 = 0001, ctr_2 = 0010, ctr_3 = 0011, etc. [2 marks]

v. How would you encrypt the 10-bit message M' = `0011 0011 11` using E_K in CTR mode? [2 marks]

vi. Define what error propagation means in a mode of operation. [3 marks]

vii. Do either ECB or CTR mode suffer from error propagation? Justify your answer. [2 marks]

(e) Draw how CBC-mode of encryption works, and describe an attack on this mode of operation when non-random IVs are used. [5 marks]

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Question 2 - Provable Security

(a) How is a security notion typically defined in cryptography? [2 marks]

(b) State the three resources required by an attacker that are usually considered when establishing the complexity of an attack. [3 marks]

(c) Describe the attack complexity of *exhaustive key search* given a plaintext/ciphertext pair (p, c) such that c is the symmetric encryption of p under some secret key k. [6 marks]

(d) Provide a description for the IND-CPA game for symmetric encryption schemes. [7 marks]

(e) How does the IND-CPA game for public-key encryption differ from the one in the symmetric setting? [3 marks]

(f) Does textbook RSA encryption scheme satisfy IND-CPA? Justify your answer. [2 marks]

(g) The security notions in part (d) and (e) of this question model indistinguishability of ciphertexts, i.e., encrypted messages. In the context of public-key encryption, it is meaningful to consider *indistinguishability of encryption keys*, and its corresponding security game.

   i. Provide a description for a security game in which the adversary's goal is to distinguish the public-key encryption keys. [8 marks]

   ii. What security property would such a game capture? [3 marks]

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Question 3 - Public-key Cryptography

(a) Define the Discrete Logarithm Problem (DLP), and name two algorithms that can be used to solve it. [4 marks]

(b) The Diffie-Hellmann Key Exchange protocol.

i. Describe in detail the protocol, specifying the setup, the parties involved, the exchanged messages and the final output. [6 marks]

ii. What problem is the security of the Diffie-Hellmann protocol based on? [2 marks]

iii. How does the problem identified in part ii) relate to the DLP? [2 marks]

(c) Schnorr Signatures.

i. Describe the Schnorr digital signature scheme, detailing the key generation, signing and verification algorithms. [6 marks]

ii. Identify what the typical parameter sizes for this signature scheme are. [4 marks]

iii. Is it safe to re-use randomness in this scheme? Justify your answer. [4 marks]

(d) Define *post-quantum cryptography* and give two examples of areas in cryptography that classify as post-quantum. [6 marks]

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Question 4 - Applications of Cryptography

(a) Describe the main security goals of the TLS Handshake Protocol. [3 marks]

(b) The TLS Handshake Protocol has *ciphersuite negotiation* as a feature. Explain what this feature does, why it is important, and the mechanism by which it is performed in TLS. [4 marks]

(c) The TLS Handshake Protocol provides a number of different mechanisms for establishing keys. One of them uses RSA-based public-key encryption, which until recently was the most popular choice.

i. Describe this specific mechanism, focussing in your answer on the message exchanges that take place between the protocol participants to negotiate and then use this mechanism. If you wish, you may use a diagram to illustrate your answer. [6 marks]

ii. The use of RSA-based public key encryption in TLS, in combination with the `ServerFinished` message, also provides the basis of authentication of the server to the client in the TLS Handshake Protocol. Explain why. [3 marks]

iii. The more recent TLS 1.3 standard no longer supports RSA-based key establishment due to its lack of *forward security* (or *secrecy*). Explain why RSA-based key establishment does not offer forward secrecy, and why the lack of forward secrecy may be a considered a vulnerability in key establishment protocols. [2 marks]

(d) State two security properties the TLS Record Protocol provides. [4 marks]

(e) Explain what AEAD is and how it is used in the context of the TLS Record Protocol. [8 marks]

(f) Discuss the threats (if any) that scalable and reliable quantum computers would pose to the security of current versions of the two TLS subprotocols: the TLS Handshake Protocol and the TLS Record Protocol. [4 marks]

END

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